

PANEL: GENERATIVE AI FOR CHIP DESIGN: ARE WE THERE YET?

GenAI for Chip Design Applications and Challenges

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Agent for Chip Design Applications

Coding Assistance → Generating design and verification code (RTL, SVA, testbenches, EDA scripts, tools scripts, and configs)

- Generate code from specifications

- Generate code for auxiliary design tasks such as assertions, comments, etc.

- Generate lower-level programs from higher-level descriptions

- Generate scripts for specific tasks (VLSI, Verification)

- Transform code for more efficient implementation

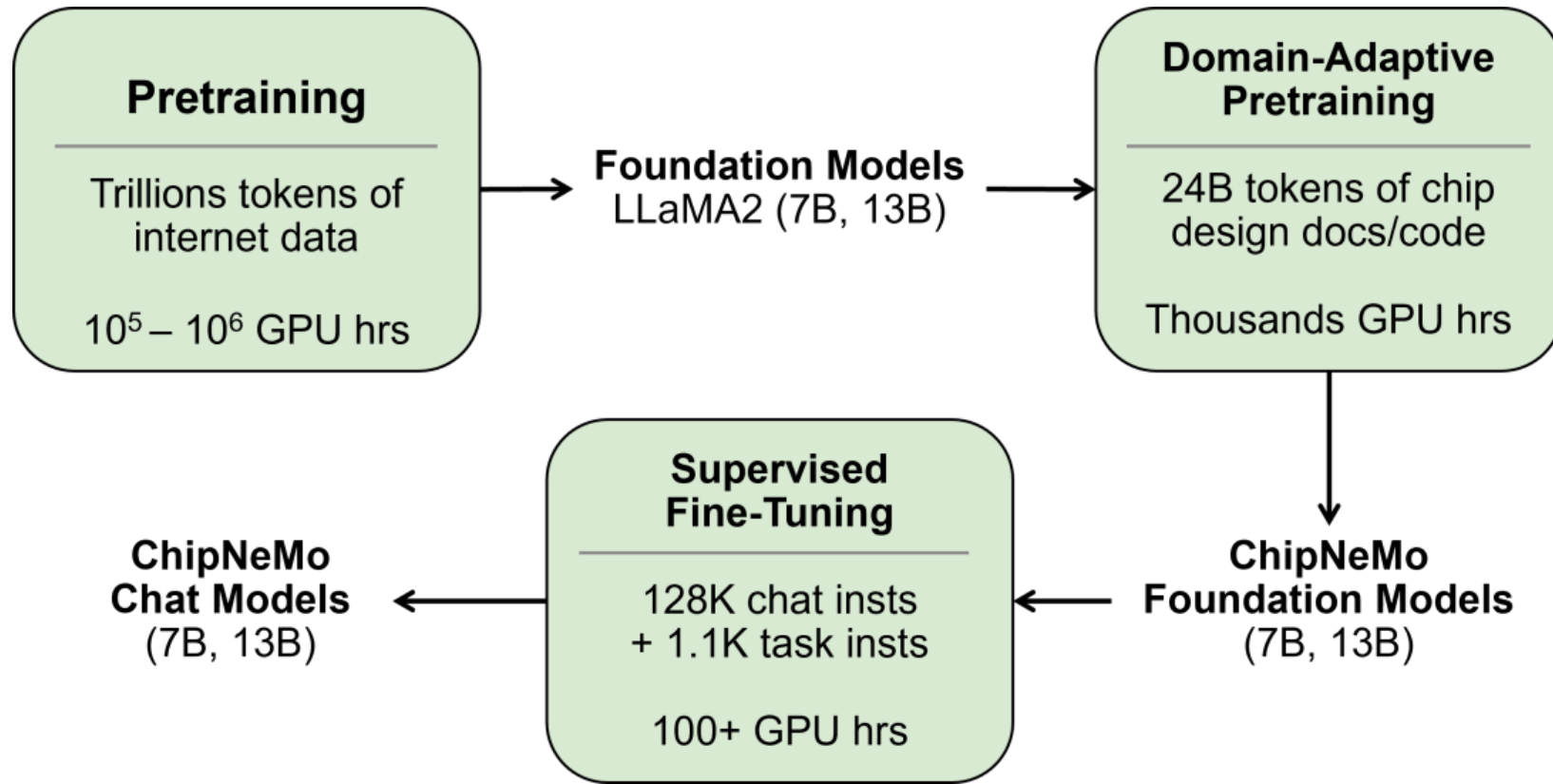
Know-how Assistance → : Generating insights, knowledge, and ideas. Answer questions about designs, infrastructures, tools, flows, HW domains, etc.

Analysis Assistance → Summarization, report/log analysis, visualization of design and related data, check rule violations, etc.

Debug Assistance → Triage and fix a design problem, e.g. regression failures for logic design and timing failures for physical design.

Optimization Assistance → Optimize for design and verification: coverage closure, recipe optimization, architecture exploration, etc.

Domain Adapted Models



ChipNeMo 2023

RTL Design Benchmarking

Question Prompt:

Implement the Verilog module based on the following description. Assume that signals are positive clock/clk edge triggered unless otherwise stated.

Problem Description:

Given an 8-bit input vector [7:0], reverse its bit ordering.

```
module top_module (
    input [7:0] in,
    output [7:0] out
);
```

Canonical Solution:

```
assign {out[0], out[1], out[2], out[3], out[4],
        out[5], out[6], out[7]} = in;
endmodule
```

VerilogEval (2023)

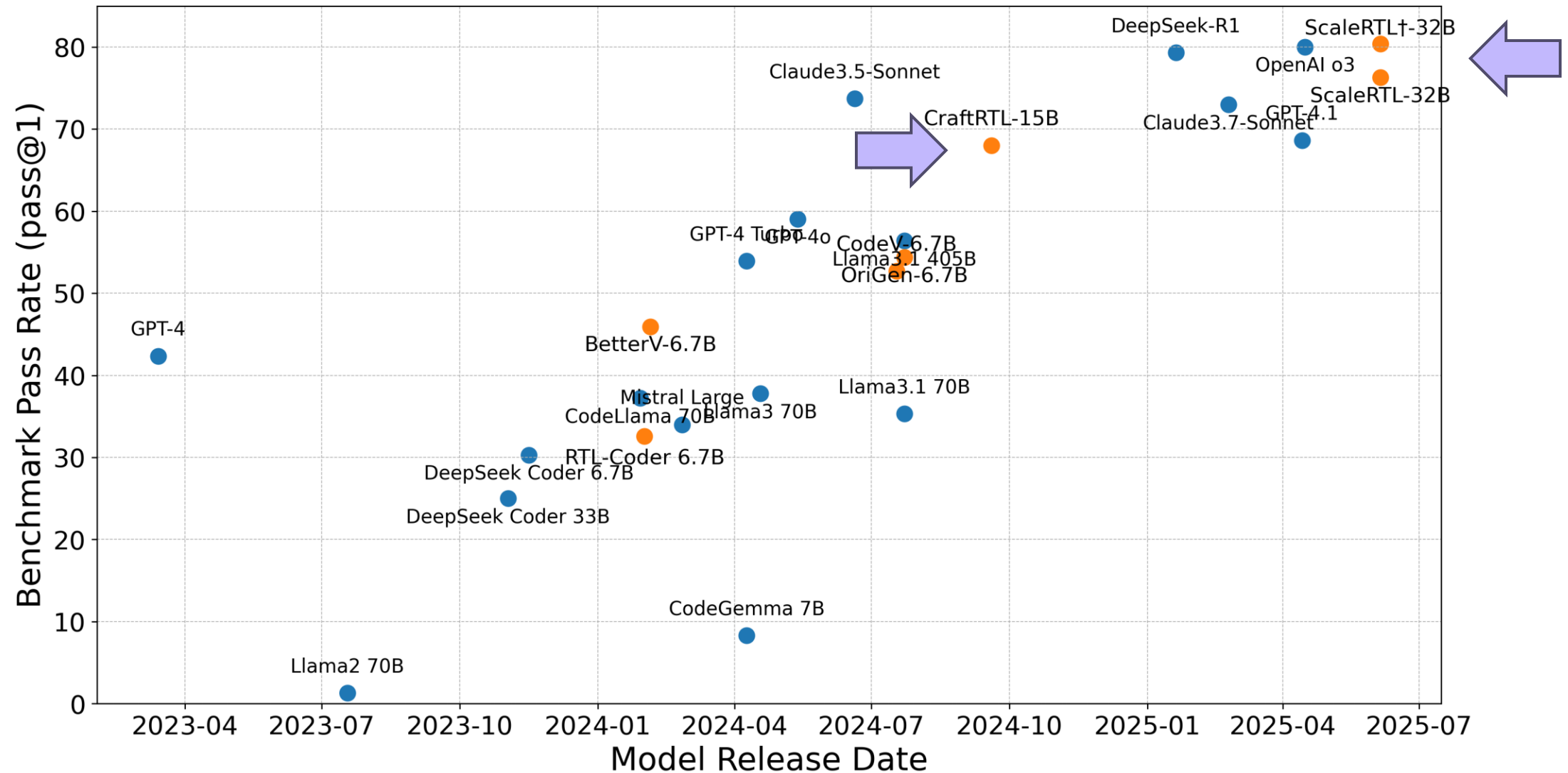
70+% pass rate

ID	Category Description	Volume		Tokens Mean/Max (k)	
		Non-Agentic	Agentic	Non-Agentic	Agentic
Code Generation					
cid02	RTL - Code Completion	94	0	1.5/4.5	–
cid03	RTL - Natural Language Spec to Code	78	37	1.2/6.9	2.7/7.9
cid04	RTL - Code Modification	56	26	2.0/4.6	5.7/19.5
cid05	RTL - Spec to RTL (Module Reuse)	0	26	–	7.4/28.5
cid07	RTL - Code Improvement (Linting/QoR)	41	0	1.9/5.9	–
cid12	Design Verification - Testbench Stimulus Gen.	68	18	1.4/6.2	2.1/4.6
cid13	Design Verification - Testbench Checker Gen.	53	18	2.8/7.3	4.5/10.7
cid14	Design Verification - Assertion Generation	68	30	2.6/7.5	4.8/14.6
cid16	Design Verification - Debugging / Bug Fixing	36	11	2.3/6.5	3.9/14.5
Code Comprehension					
cid06	Correspondence - RTL to/from Specification	34	0	1.6/5.5	–
cid08	Correspondence - Testbench to/from Test Plan	29	0	3.1/6.1	–
cid09	Question & Answer - RTL	34	0	1.1/5.0	–
cid10	Question & Answer - Testbench	26	0	3.6/4.8	–
	Total # of Problems	617	166		

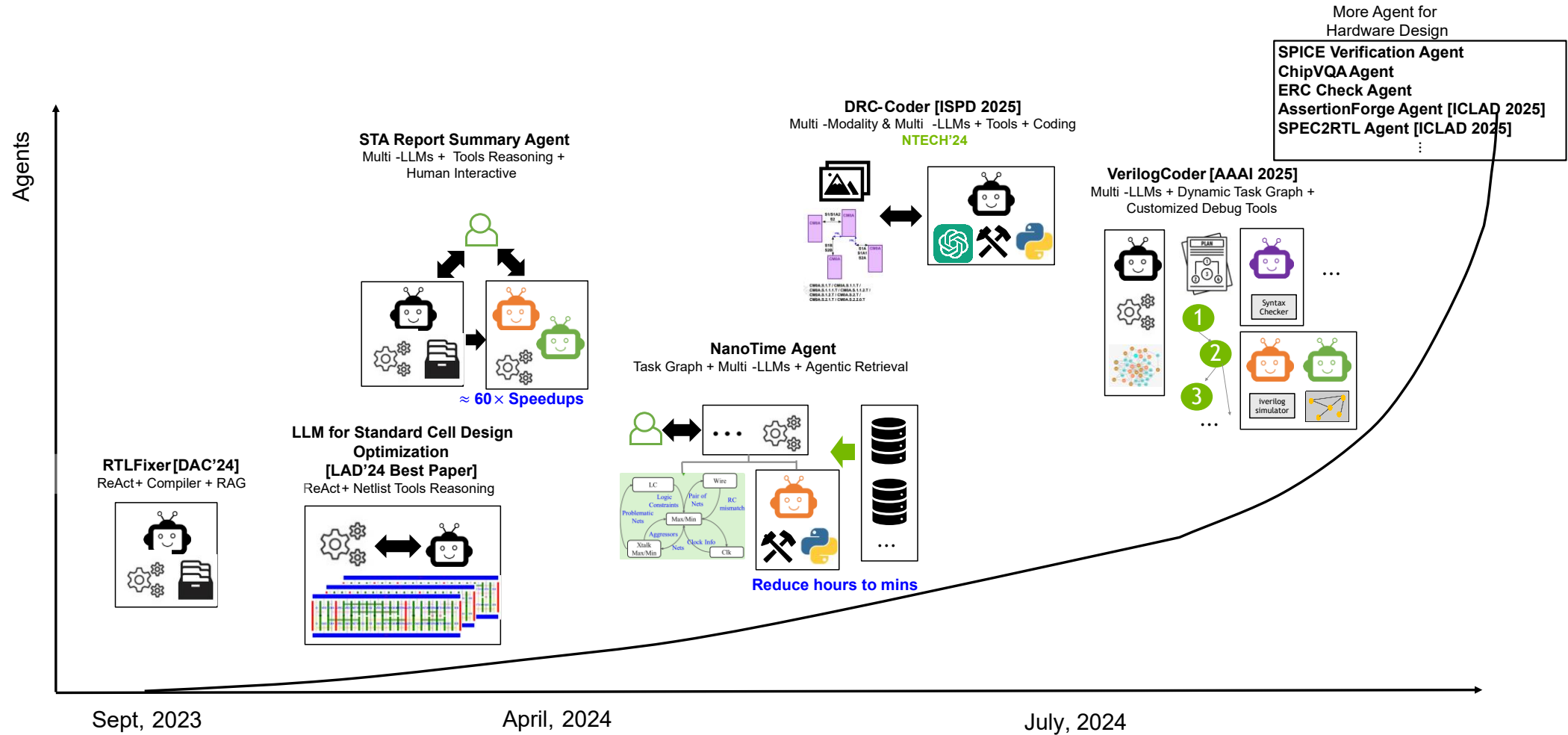
CVDP (2025)

10%-40% pass rate

RTL Models



Evolution of Custom Agents for Hardware Design



Challenges and Opportunities

How to cope with large legacy code base?

How to re-define arch/micro-arch spec for AI to generate code

How to effectively program domain-specific languages used in hardware design?

How to effectively teach LLM/Agent domain knowledge?

How to effectively use existing EDA tools – Agentic EDA

Should EDA tools change since users are Agents instead of human?

How to modify design methodology to make it suitable for AI instead of Human?

How to effectively organize teams to cowork with Agents

